

Simple agent-based models of markets

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Financial markets as complex systems

Some stylized facts:

- price is generated endogenously
- agents are not fully rational (herding behavior)
- distribution of returns is power-law tailed (not gaussian!)
- it gets more gaussian as the time-scale of returns is increased
- volatility is positively autocorrelated over timescales of days (volatility clustering)

Herding behavior: advantages and limits

Simple model to understand the (excessive) role of herding behavior.¹

Consider N agents facing binary choices (buy/sell). Two types of agents (fixed strategies):

- **fundamentalists**: process information and analyse the market based on fundamental values. They receive a private random signal θ , which is right with probability $p > 0.5$.
- **chartists**: observe the actions of the others (technical analysis) and decide (trend-followers)

There is a fraction z of fundamentalists (resp. $1 - z$ of chartists).

Dynamics: at each time step t , a chartist adopts the majority opinion observed at time $t - 1$ in a randomly selected group of m agents. Chartists initially play randomly.

¹P. Curty and M. Marsili, *JSTAT* (2006)

Herding behavior: advantages and limits

Define:

- Probability q_t that a chartists makes the right choice at time t
- Probability π_t that a random agent makes the right choice at time t

$$\pi_t = zp + (1 - z) q_t$$

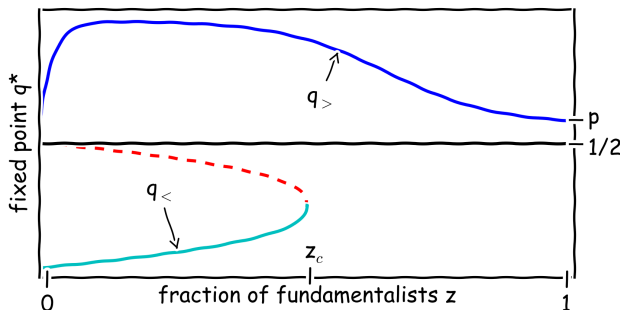
Because of the random sampling by chartists

$$q_{t+1} = \sum_{\ell=(m+1)/2}^m \binom{m}{\ell} \pi_t^\ell (1 - \pi_t)^{m-\ell} \equiv F_z [q_t]$$

In the long run, we expect a fixed point $q_t \rightarrow q^*(z)$, $\pi_t \rightarrow \pi^*(z)$ such that

$$q^* = F_z [q^*]$$

Herding behavior: advantages and limits



- for $z > z_c$: only one (attractive) fixed point exists, $q_> > p$. Information aggregation works! Herding is favorable. The success of trend-followers increases with their number.
- for $z < z_c$: three solutions (two are stable). The solution $q_>$ becomes very bad (random prediction) if $z \rightarrow 0$. The solution $q_<$ is always even worse than random prediction. Information aggregation doesn't work!

Herding behavior: advantages and limits

What if we let agents myopically self-organize changing strategy?

- Since information retrieval has a cost, herding is chosen as long as $\langle q_t \rangle > p$, that makes z further decrease.
- However, when $\langle q_t \rangle < p$, agents will prefer become fundamentalists, then z further increases.

Nash equilibrium: fraction z_{NE} at which both strategies are equally successful ($\langle q_t \rangle = p$).

Notice that $z_{NE} = O(1/\sqrt{N}) \rightarrow 0$ for $N \rightarrow \infty$. Herding behavior dominates the market!

Minimal agent-based model of a market

We have seen why markets can coordinate on the wrong information based on herding behavior.

How do prices and trading activity fluctuate when agents switch between information-based and imitation-based strategies?

A self-consistent mis-belief leads to self-consistent mispricing, that implies self-organization towards a regime of volatility clustering, intermittency, price bubbles, ...

Let's build a minimal model of a **market as a self-organizing complex system**.

Minimal agent-based model of a market

The Lux-Marchesi model² is a minimal agent-based model of financial markets that reproduces key empirical stylized facts of asset returns:

- N agents choosing between two strategies: fundamentalists (n_f) and chartists (n_c)
- chartists can be optimistic (n_+) or pessimistic (n_-) about price dynamics
- agents adapt their strategy based on utility
- real price dynamics is endogenous, related to the excess demand
- fundamental price value dynamics is exogenous

²T. Lux and M. Marchesi, *Nature* (1999)

Minimal agent-based model of a market

Transitions within chartists:

$$\begin{aligned}W[- \rightarrow +] &= v_1 \frac{n_c}{N} e^{U_1} \\W[+ \rightarrow -] &= v_1 \frac{n_c}{N} e^{-U_1}\end{aligned}$$

with utility

$$U_1 = \alpha_1 \frac{n_+ - n_-}{n_c} + \frac{\alpha_2}{v_1} \frac{d}{dt} \log p$$

v_1 measures the typical frequency of agents' revision of expectations.

Minimal agent-based model of a market

Transitions between chartists and fundamentalists

$$\begin{aligned} W[f \rightarrow +] &= v_2 \frac{n_f}{N} e^{-U_{2,1}} & W[+ \rightarrow f] &= v_2 \frac{n_+}{N} e^{U_{2,1}} \\ W[f \rightarrow -] &= v_2 \frac{n_f}{N} e^{-U_{2,2}} & W[- \rightarrow f] &= v_2 \frac{n_-}{N} e^{U_{2,2}} \end{aligned}$$

with

$$\begin{aligned} U_{2,1} &= \alpha_3 \left[\left(\frac{r}{p} + \frac{1}{v} \frac{d}{dt} \log p - R \right) - s \left| \frac{p_f - p}{p} \right| \right] \\ U_{2,2} &= \alpha_3 \left[- \left(\frac{r}{p} + \frac{1}{v} \frac{d}{dt} \log p - R \right) - s \left| \frac{p_f - p}{p} \right| \right] \end{aligned}$$

First terms represent the profit (nominal dividends and capital gain due to price change) of chartists from the two groups, the second term is the time-delayed discounted ($s < 1$) profit of fundamentalists.

Average real return by holders of assets is $R = r/p_f$.

Minimal agent-based model of a market

Price change: endogenous response of the market to the imbalance between demand and supply.

Excess demand of chartists: $ED_c = (n_+ - n_-) / t_c$

Excess demand of fundamentalists: $ED_f = n_f \gamma (p_f - p) / p$

where t_c is the average trading volume per transaction

Price transitions occurs with probability:

$$W[p \rightarrow p + \epsilon p] = \max(0, \beta (ED_c + ED_f + \mu))$$

$$W[p \rightarrow p - \epsilon p] = -\min(0, \beta (ED_c + ED_f + \mu))$$

that is a discretized stochastic version of the ODE

$$\dot{p} = \beta (ED_c + ED_f) p$$

Fundamental price changes as

$$\log p_{f,t} - \log p_{f,t-1} \sim \mathcal{N}(0, \sigma)$$

Minimal agent-based model of a market

Mean-field approximate ODE analysis:

$$\begin{aligned}\frac{dx}{dt} &= 2\eta v_1 (\tanh(U_1) - x) \cosh(U_1) \\ &\quad + (1 - \eta) (1 - x^2) v_2 [\sinh(U_{2,1}) - \sinh(U_{2,2})] \\ \frac{d\eta}{dt} &= (1 - \eta) \eta (1 + x) v_2 \sinh(U_{2,1}) + (1 - \eta) \eta (1 - x) v_2 \sinh(U_{2,2}) \\ \frac{dp}{dt} &= \beta \left(x \eta t_c + (1 - \eta) \frac{p_f - p}{p} \gamma \right) N p\end{aligned}$$

with $x = (n_+ - n_-) / n_c$ and $\eta = n_c / N = (1 - z)$

Stationary points:

- 1 $x^* = 0, p^* = p_f$ (arbitrary η)
- 2 $x^* = 0, \eta^* = 1$ (arbitrary p)
- 3 $\eta^* = 0, p^* = p_f$ (arbitrary x)

Minimal agent-based model of a market

If we exclude absorbing states (2-3), the **fundamental stationary state** (1) is characterized by balance sentiment and price equal to the fundamental one!

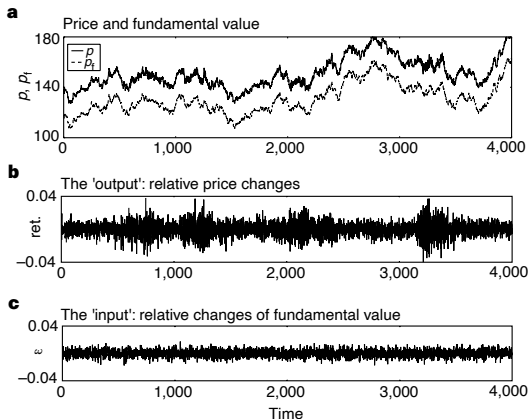
Stability analysis: the fundamental stationary state becomes unstable when some (complicated) conditions are met. It exists a threshold fraction η_c of chartists such that instability arises when $\eta > \eta_c$, i.e. for large number of chartists!

Temporal deviations from the stability region determine:

- intermittent behavior
- clusters of volatility
- fat tails in the distribution of price returns

Minimal agent-based model of a market

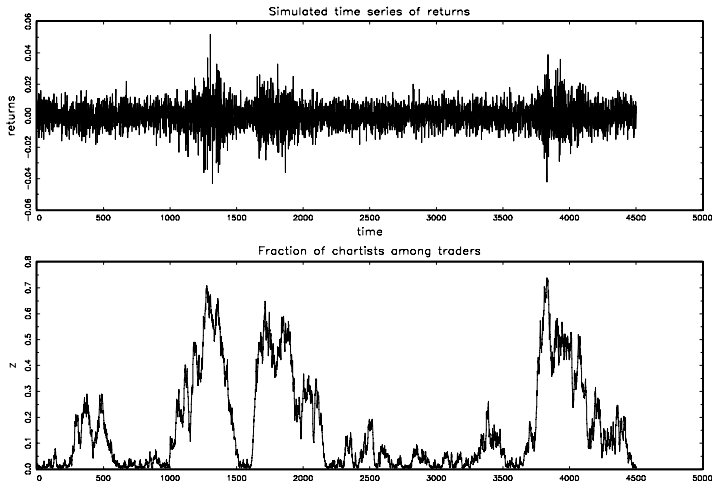
Stable stationary behavior is like a laminar flow:



Real price and fundamental price are just shifted for readability.

Minimal agent-based model of a market

Turbulent regimes where the unstable regions is visited (burst of activity, more herding):



Minimal agent-based model of a market

Probability of returns shows fat-tails at short time-scales (volatility clustering):

